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Translation of the Diabetes Prevention Program to Ethnic Communities in the United States

Daniel L. Hall¹ · Emily G. Lattie¹ · Judith R. McCalla¹ · Patrice G. Saab¹

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Abstract The Diabetes Prevention Program (DPP), an evidenced-based lifestyle intervention for type 2 diabetes (T2D), has been translated for use with ethnic minority communities throughout the United States that are disproportionately at-risk for T2D. The present paper sought to critically review ethnic translation studies of the DPP with respect to translation methods utilized, the success of these methods, and alternative or supplemental methodologies for future translation efforts. Manuscripts reviewed were found by searching PubMed and PsycINFO, using the terms: “diabetes prevention program” AND [“translation” or “ethnic”]. Of 89 papers found, only 6 described ethnic translations of the DPP in the United States, and were included in this review. Translations of the DPP to African American, Hispanic/Latino, Native Hawaiian and Other Pacific Islander, Arab American, and American Indian and Native Alaskan communities were identified and reviewed. The most common translation strategies included group-based delivery and use of bilingual study personnel. Generally, these factors appeared to increase acceptability of the intervention within the ethnic communities reviewed, and should be considered in future efforts to implement and translate the DPP to ethnic communities in the United States.

Keywords Diabetes prevention · Lifestyle intervention · Translational research · Minority health

Numerous non-pharmacological interventions have been developed with the aim of preventing the incidence of type 2 diabetes (T2D). Among the most promising are those targeting diet/nutrition and physical activity in individuals deemed at-risk. Several lifestyle interventions have demonstrated high efficacy, such as the Malmö Study, the Finnish Diabetes Prevention Study, the China Da Qing Diabetes Prevention study, and the Indian Diabetes Prevention Programme [1–4]. These international interventions, while efficacious, have spawned few if any implementation efforts targeted to and/or tailored for ethnic minority groups in the United States. Alternatively, the Diabetes Prevention Program (DPP), another prominent lifestyle intervention for T2D conducted in the United States, has influenced the implementation of programs for use with specific ethnic minority communities. This review will focus on efforts to translate the DPP for specific ethnic groups, many of which are disproportionately affected by T2D. Promising translation studies will be critically reviewed in terms of (a) translation methods utilized, (b) the success of these methods, and (c) alternative or supplemental methodologies for future translation efforts.

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Background

Type 2 diabetes is categorized by a resistance to insulin, which can arise from both genetic and environmental influences [5, 6]. It is a leading cause of death in most developed nations, a risk factor of many chronic illnesses including cardiovascular disease and cancer, and is

associated with diminished quality of life across a number of domains [7–9]. While approximately 24 million people in the United States have T2D, far more have prediabetes, i.e., higher-than-normal blood glucose levels, placing them at high risk for T2D development [10]. Most recent national estimates indicate that 35 % of adults the United States have prediabetes; this equates to approximately 79 million Americans [5]. Importantly, United States public health data indicate that ethnic minorities are disproportionately at-risk for developing T2D [11, 12]. Ethnic communities in the US with higher prevalence rates of T2D than non-Hispanic whites (7.1 %) include African Americans (12.6 %), Hispanics/Latinos (11.8 %), Native Hawaiians and Other Pacific Islanders (22.7 %), Asian Americans (8.4 %), and American Indians/Native Alaskans (16.1 %) [13].

Modifiable lifestyle factors such as physical activity level and diet are reliable predictors of T2D prevention [14]. In terms of physical activity, a sedentary lifestyle is consistently a robust predictor of T2D onset in both men and women [15, 16]. In terms of diet, meta-analyses of diet-focused interventions for those at risk of T2D have identified the following factors that lower odds of disease onset: low intake of saturated fats, total fat, and trans fatty acids as well as high intake of non-starch polysaccharides, omega-3 fatty acids, and low glycemic index foods [12, 17].

The DPP: Intervention and Evidence

The DPP, a product of the Diabetes Prevention Program Research Group, demonstrated high efficacy in a randomized clinical trial involving 3234 prediabetic adults across 27 centers in the United States [18]. Participants were randomized into one of three groups: a placebo control group, a metformin (pharmacological agent) group, and a lifestyle intervention group. The lifestyle intervention group aimed to reduce participants' weight by a minimum 7 % and engage participants in moderate physical activity at least 150 min per week. To achieve this goal, lifestyle intervention-randomized subjects participated in an individually delivered protocol [18]. A 16-week core, comprising the initial sessions, included education about nutrition and exercise benefits and techniques, behavioral support (via individual counseling on self-monitoring, goal setting, discussion of barriers, and problem solving strategies for overcoming barriers, stimulus control, and relapse prevention training), and motivational interviewing. Interventionists were case managers with training in health promotion. Participants were offered supervised group exercise classes twice per week, in addition to group maintenance courses, lasting 4–6 weeks, offered 4 times per year. If a participant was not meeting their goal, a “tool box” of problem solving strategies was offered.

Overall, the DPP lifestyle intervention resulted in a reduced risk (by 58 % relative to placebo) of developing T2D at 2 2/3 year follow-up [18]. On average, participants in the intervention group had greater weight loss and physical activity than the placebo control or metformin groups [18]. Effects were maintained at 10-year follow-up [19]. Importantly, although the sample was ethnically diverse (only 45 % non-Hispanic white), treatment effects did not differ by ethnic group, providing an impetus for the National Institutes of Health and the US Department of Health and Human Services to fund research establishing the translatability of the DPP in communities comprised of at-risk ethnic groups. Considering the high level of resources used by the DPP, establishing its effectiveness after translation to community settings is a crucial step in understanding its potential for implementation into policy and practice [20].

Methods

Recently, Whittemore [21] evaluated the reach, efficacy, adoption, implementation, and maintenance of 16 translations of the DPP across various medical and community settings. Due to its broad scope, the review reported samples' levels of diversity (which ranged from 0 to 100 % non-Hispanic whites) yet did not offer critiques of ethnicity-focused translation methods and outcomes. Translations were identified by Whittemore that targeted African Americans [22], Hispanic/Latino-Americans [23], and Native Hawaiians and Other Pacific Islanders [24]. Additional studies targeting Arab Americans [25], American Indians/Native Alaskans [26], and African Americans [27] have since been published. To date, no study has reviewed these ethnically-targeted translations of the DPP with respect to culturally-relevant translation methods utilized, the success of these methods, and alternative or supplemental methodologies for future translation efforts.

Manuscripts considered for this review were found by searching PubMed and PsycINFO, using the terms: “diabetes prevention program” AND [“translation” or “ethnic”]. Of 89 papers found, only 6 described ethnic translations of the DPP in the United States, and were included in this review. These studies are reviewed below and summarized in Table 1.

Ethnically-Translated DPP Studies

African American Translation

Among ethnic minorities in the United States, African Americans have the fastest increase in T2D incidence in

Table 1 Summary of ethnic translations of the DPP

Study	Population	Translation methods utilized	Success of methods	Suggestions for future translation efforts
Boltri et al. [22, 27]	African American	Group delivery; tested in 16 session and condensed 6 session versions; conducted in Baptist churches; sessions began with prayer	Variable attendance (65 % in the 2008 paper, 62.1 % in the 2011 paper); significant reductions (3.6 %) in weight, blood pressure, BMI, and fasting glucose levels at 6 and 12 months; participants developed a faith-based maintenance program indicating continued interest	Target high-fat food choices by modifying the use of butter and frying food preparation techniques; validate the significance of meals as social events; use demonstration, story-telling, and hands-on methods when teaching healthier food preparation; emphasize the increased risk of T2D development in African Americans; emphasize group exercise techniques; emphasize the value “determination”
Merriam et al. [23]	Hispanic/Latino	Bilingual interventionists; partnerships with Spanish and Latino-utilized community resources for recruitment; diet modifications for traditional Latino diet; written materials were “low” literacy and heavy use of visuals	Retention at a 1-year follow-up was high (7 % attrition); all participants chose to use Spanish language materials; participants had high levels of trust in the intervention and study personnel	Target health information and cognitive therapy techniques such as reframing and addressing maladaptive beliefs about exercise; emphasize resistance training; incorporate music, song, dance, and being outdoors for physical activity
Mau et al. [24]	Native Hawaiians and Other Pacific Islanders	Equal partnership of Community Investigators and Academic Investigators; use of information from focus groups of community members and informant interviews of community leaders; protocol cut to eight sessions over 12 weeks; delivered in groups led by community peer-educators; added information on the economics of eating healthy and on effective communication with doctors	Participants lost average of 1.5 % of their body weight at 12 weeks; mean diastolic and systolic blood pressures were slightly lower at 12 weeks; 6-min walk test distance improved at 12 weeks	Utilize monthly group booster sessions; engage individuals within their existing social networks, such as church groups, and emphasize faith-based values
Jaber et al. [25]	Arab American	Focus groups helped generate topic areas for intervention: gender-specific groups (of 10–12 participants) for education and physical activity sessions, emphasizing women’s role as health promoter in the home, enhancing male participation in food preparation, and incorporating Arabic sayings and religious themes; bilingual study personnel for all study protocol delivery; individuals not initially interested in participating were invited for four-session seminar targeting ambivalent participants’ beliefs about diabetes	92 % of participants attending at least 10 weekly sessions; significant reductions in total and saturated fat intake; 75 % achieved the 150-min physical activity goal; 49 out of 63 participants initially uninterested enrolled after 4 session seminar	Determine feasibility of translation for less-knowledgeable or non-Arab American interventionists to implement in other Arab American communities or healthcare/community clinics. Utilization of control group
Jiang et al. [26]	American Indian and Native Alaskan	Strong partnership across 36 AI/AN health care programs; each program ran 16 session Life Balance Curriculum from the DPP in group settings in a 16–24 week time period but adaptations to the curriculum for local cultures was allowed; familiar AI/AN traditions, such as talking circles, indigenous foods, and drumming, were incorporated	Participants attended average of 13/16 classes; those who finished all 16 classes had substantially lower diabetes incidence (at 3.5 % per year) than participants who did not finish all of the classes (at 7.5 % per year); significantly higher levels of physical activity post-treatment (~181 min/week) compared to baseline (~99 min/week) across centers; average weight loss between baseline and post-treatment assessment was 4.4 %; fasting blood glucose decreased from baseline to follow-ups	Additional focus on developing effective retention strategies such as collecting contact information for other family members, and sending text messages and e-mails to contact participants who may be hesitant to answer the telephone; utilization of videoconferencing as a method for treatment delivery

recent years [11, 13]. It is no surprise then that diabetes is a major health concern for African Americans, stemming from increased knowledge about T2D and its increasing visibility within tight-knit faith-based communities [28]. Indeed, faith-based community interventions may be particularly salient for African Americans, as they have high rates of church attendance [29]. Thus, a faith-based adaptation of the DPP for African Americans at risk of T2D was developed Boltri et al. [22, 27] aiming to reduce weight, improve blood pressure, and stabilize fasting glucose levels among participants. After screening for pre-diabetes (operationalized in this study as fasting glucose levels of 100–125 mg/dL), eight congregants were eligible and participated in the 2008 study, while 37 were eligible and participated in the 2011 study.

Translation Methods Utilized

The African American adaptation made several changes to the original DPP protocol. The intervention tested both a 16-session protocol [22], reflecting the core DPP program, as well as in a 6-session protocol with condensed sessions [27]. Follow-up or booster sessions were not conducted.

Intervention materials were modified for group delivery, and questions were reportedly added to facilitate group discussion. The intervention delivery occurred in five African American Baptist churches in Macon, Georgia. The pastor's approval and input was solicited by the researchers, and recruitment involved announcements in Sunday bulletins as well as a research member explaining the DPP briefly during church service. A deacon or minister led the group in a prayer at the start of each session. The content and length of these prayers was not reported, but one may assume these prayers mentioned the Christian faith and/or physical health.

Success of Translation Methods

The authors did not report measuring acceptability of the intervention by participants, or provide commentary on subjective feedback about the translated protocol. However, they noted that participants developed a faith-based maintenance program after the study ended, which suggests that at least some participants maintained their interest in the intervention [27]. Overall, attendance was variable (65 % in the 2008 paper, 62.1 % in the 2011 paper), and all participants completed follow-up measurements of outcomes at 6 and 12 months. Despite a small sample, significant reductions (3.6 %) in weight, blood pressure, BMI, and fasting glucose levels were evident at 6 and 12 months post-intervention.

Alternative or Supplemental Methodologies for Future Translation Efforts

The translations by Boltri et al. [22, 27] provided preliminary evidence for a faith-based adaptation of the DPP administered to groups in a church setting. The intervention was targeted to church attenders, as group prayers were an integral part of sessions. However, the intervention was also targeted to African Americans, and there was no mention of modifications made for this specific ethnic group. For instance, Boltri et al. [22, 27] did not consider culturally competent methods for promoting healthy adaptations to a traditional African American diet which have been successful in other studies [30]. These approaches target high-fat food choices by modifying the use of butter and frying food preparation techniques [31]. Additionally, the investigators could have targeted *delivery* of diet modification aspects of the interventions. An interventionist could validate the significance of meals as social events, and use demonstration, story-telling, and hands-on methods when teaching healthier food preparation, as these cultural components may be highly relevant for African Americans [30]. Researchers could also follow the suggestion of Hawkins et al. [32] and personalize the DPP by emphasizing the increased risk of T2D development in African Americans. For increasing physical activity, group exercise techniques could be emphasized, as African Americans may be encouraged by the social nature of group-based physical activity [33]. The value “determination” could also be reinforced to encourage physical activity in this population [33].

Hispanic/Latino-American Translation

Hispanics/Latinos comprise the largest ethnic minority in the United States [34]. The incidence rate of T2D among Hispanics/Latinos is alarmingly high, with approximately one in two Hispanics/Latinos in the United States developing T2D during their lifespan [35]. An adaptation of the DPP was developed to ameliorate the risk of T2D in this at-risk group, with similar outcome aims as the original DPP intervention [23]. The randomized clinical trial had two groups: usual care or the translated lifestyle intervention. Recruitment and intervention delivery were located in the mostly Latino community of Lawrence, Massachusetts. The study has not yet published results, other than baseline characteristics of the 312 enrolled participants [23]. Follow-up occurred at 1 year post-recruitment, and will continue for 7.5 years.

Translation Methods Utilized

Merriam et al. [23] dedicated an entire manuscript to detailing their intervention methodology. Translation strategies broadly fell into three categories: research staffing, recruitment, and intervention tailoring. The study employed bilingual interventionists (personal trainers, dietitians, and other study staff), which allowed participants to engage with the intervention in their native language.

Recruitment was targeted to Latinos and tailored to their community, both in use of Spanish and Latino-utilized community resources. Partnering with multiple community sites such as the local YWCA, a town family health center, and a local aging/senior center, the study utilized several types of media, including Spanish television, radio, and newspaper advertising, as well as bilingual fliers and mailings to community centers and local physicians.

In addition to these recruitment strategies, Merriam and his research team, several of whom are Latino/a, adapted the intervention to reflect cultural competence. Diet modifications were targeted to address a traditional Latino diet, characterized by a focus on high sodium, saturated fat, hydrogenated fat, large portions, and refined carbohydrates and starches. Participants were encouraged to eat frequent meals of smaller portion sizes and to use low-fat and high-fiber foods. The interventionist taught healthy cooking, food shopping, and problem-solving skills in a hands-on fashion, encouraging participants to engage with study materials and practice problem-solving techniques in-session. Written study materials were developed using “low” literacy levels and heavy use of visuals. One example is the use of a food guide based on a Latino diet; healthy foods were marked in green, questionable foods in yellow, and unhealthy items marked in red. Additionally, a manual was created for interventionists to aid in the implementation of the culturally-adapted protocol.

Success of Translation Methods

While results of the study have yet to be published, the authors reported preliminary acceptability data about their trial [23]. Retention at a 1-year follow-up was high (7 % attrition). Additionally, all participants chose to use Spanish questionnaires, indicating that language was an important adaption among their sample of Hispanic/Latino Americans. Participants reportedly had high levels of trust in the intervention and study personnel. The authors posit this trust was a product of using established community centers and leaders, in addition to using Spanish-speaking staff. The decision to use bilingual or Spanish-speaking staff was well-informed for several reasons. Non-English speaking participants could participate, which overcame potential language barriers.

Additionally, the study used tools designed for oral administration to ensure that high illiteracy rates among Hispanic/Latino Americans would not preclude participants’ ability to benefit from the intervention [36]. Finally, research shows that delivery of health-related information in a recipient’s native language improves the potency of information retention and internalization [37].

The study’s translation of written materials was also important. The use of visual aids when tailoring printed materials is of high value, in part because it decreases the cognitive demand on the part of the recipient [32, 38]. Therefore, this aspect of their intervention might prove to be particularly potent. Additionally, written materials were offered in Spanish and English. Since all participants opted to use materials in Spanish, the strategy of targeting materials to participants’ native language was successful in making the intervention accessible.

Alternative or Supplemental Methodologies for Future Translation Efforts

Overall, Merriam et al.’s Hispanic/Latino translation of the DPP was impressive in its use of translational methods. It is unclear, however, the extent to which the physical activity component was translated for this population. The adapted protocol mentions emphasizing walking, but does not discuss alternative forms of exercise, barriers to exercising regularly, or problem-solving strategies that may be of relevance to Hispanics/Latinos. It has been documented that this community exercises less regularly than non-Hispanic whites, so it would be appropriate to consider ethnicity in the physical activity component of the intervention [39]. Hispanics/Latinos may have misperceptions about physical activity in terms of form and frequency, which would be ripe for targeting with health information as well as cognitive therapy techniques such as reframing and addressing maladaptive beliefs about exercise [40]. Resistance-training may be a helpful form of exercise among Hispanic/Latino males [41]. Music, song, dance, and being outdoors have also been suggested as elements to encourage physical activity in Hispanics/Latinos [33]. Future translation work could benefit from incorporating these physical activity elements in translating the DPP to Hispanic/Latino communities.

Native Hawaiian and Other Pacific Islander Translation

Strong evidence indicates that T2D prevalence is a major health concern for Native Hawaiians and Other Pacific Islanders (NHOPIs). One in five NHOPIs has T2D, and the risk of dying from T2D is three times higher than for non-Hispanic whites [42, 43]. While the DPP trial was deemed efficacious

across all included racial/ethnic groups, results were not reported specifically for NHOPIs but rather were collapsed in with all of the Asian American participants. Because T2D is a highly prevalent morbidity for NHOPIs, there is a need to develop and disseminate effective lifestyle interventions for this population. Mau et al. [24] recently reported results from a culturally-adapted translation of the DPP to NHOPIs aimed at weight reduction. The PILI (Partnerships for Improving Lifestyle) 'Ohana Project used community-based participatory research approaches to adapt the DPP to the unique concerns of NHOPIs by partnering community health organizations with researchers from the University of Hawaii.

Translation Methods Utilized

Equal partnership, of Community Investigators from the community health organizations and Academic Investigators from the University of Hawaii, was established at the beginning of the project to most effectively integrate community knowledge with scientific knowledge. Focus groups and informant interviews were conducted. Data collected from focus groups of community members and informant interviews of community leaders revealed the NHOPIs would likely benefit from specific content on the economics of eating healthy and on effective communication with doctors, and these topics were added to the curriculum. Based on the experiences of the Community Investigators in their communities who recognized a need to respect time commitments, the adapted DPP protocol was cut to eight sessions over the course of 12 weeks, and each session was scheduled to last 1.5 h or less. Groups of 6–12 individuals were run by community-peer educators rather than professionals to help breakdown community held mistrust of research and researchers [44, 45]. Mau et al. [24] noted that the curriculum underwent significant cultural and linguistic modifications to ensure that it was appropriate for the community, but specific examples of these modifications were not provided.

Success of Translation Methods

While Mau et al. [24] fully embraced a community-based, participatory research approach in adapting the DPP to fit the need of NHOPIs, results of the intervention revealed fairly modest effects. Participants completed assessments at baseline and immediately following the 12-week program, and lost 1.5 % of their body weight on average. Mean systolic (-6.00 mmHg, 95 % CI -8.5 , -3.5) and mean diastolic (-2.5 mmHg, 95 % CI -4.4 , -1.3) blood pressure measures were slightly lower at 12 weeks. Physical functioning, measured by increase in distance traveled during the 6-min walk test, also improved at 12 weeks (42 feet, 95 % CI 25, 58). The authors noted that that 26 % of

participants lost ≥ 3 % of their baseline weight, and 11 % lost ≥ 5 % of their baseline weight at the end of the 12 week trial, indicating that some participants evidenced clinically significant weight loss.

Alternative or Supplemental Methodologies for Future Translation Efforts

Mau et al. [24] identified that results for NHOPIs were not specifically examined in the original DPP study. Because the goal of this study was to adapt the DPP lifestyle intervention for weight loss, rather than diabetes prevention, individuals with a pre-existing diabetes diagnosis were included in the program. Weight loss has been shown to have some health benefits for those with diabetes, in addition to serving as a protective factor in diabetes prevention [46]. Of the 372 individuals found eligible for the study, 239 individuals participated (83 % female) and 26 % of participants reported having a diabetes diagnosis. In conjunction with the lack of a control group, the results of the Mau et al. [24] study cannot be directly compared to those from the DPP to determine if the cultural adaptation had a comparable effect for NHOPIs. While the investigators made an informed decision to limit the number of sessions and the length of the program in attempts to maximize participation, the 12-week period may not have been long enough to capture health improvements. Additionally, the lack of a longer-term follow-up does not allow researchers to determine if the lifestyle changes that were made in the group were sustained. Future work with NHOPIs may benefit from the utilization of monthly group booster sessions to allow for a built-in brief assessment and an opportunity for group members to reunite with one another and discuss barriers to their new healthy lifestyle behaviors. Sessions in the Mau et al. [24] study embraced the role of family, friends, and community members in the process of individual lifestyle change. Further work with NHOPIs may benefit from engaging individuals within their existing social networks, such as church groups, and emphasizing faith-based values, as these methods have been successful in recruitment and retention for Native Hawaiian populations [47].

Arab American Translation

There are approximately 1.67 million Arab Americans in the United States today, comprising 0.5 % of the total US population [34]. Arab Americans are unique in their cultural history and values, with religious orientation, family values, gender, language, and acculturation requiring attention from culturally competent intervention translations [25]. They are also at high risk of T2D; as many as one in four Arab Americans are prediabetic [48]. To address this public health

concern, Jaber et al. [25] developed a community-based, non-randomized intervention translating the DPP intervention for use with Arab Americans. Participants were self-identified Arabs or Arab Americans living in Dearborn, Michigan, a city with a high Arab American concentration. After screening, 71 participants were eligible and enrolled in the intervention. The interventionists revised the length of the original DPP to 12 core sessions plus an optional session on the topic of fasting for Ramadan. In addition, optional monthly sessions were offered to discuss barriers and ways to address them. The entire intervention was delivered over 24 weeks.

Translation Methods Utilized

Jaber et al. [25] used several strategies to make the DPP culturally targeted to Arab Americans. The investigators conducted focus groups to guide modification decisions, practiced culturally-sensitive teaching methods, used a group intervention format, targeted the intervention content, utilized bilingual study personnel, and offered a “pre-intervention” educational intervention to recruit initially-ambivalent participants.

In accordance with community-based participatory research practice, the investigators first solicited input from community members to identify key intervention targets. Focus groups generated several topic areas for the translation to consider: cultural sensitivity, gender-specific groups for education and physical activity sessions, emphasizing women’s role as health promoters in the home, enhancing male participation in food preparation, and incorporating Arabic sayings and religious themes. Jaber et al. [25] emphasize that these concerns were used to gauge the cultural appropriateness of their intervention, and when necessary, to modify its content.

The authors also describe using educational styles readily accepted by members of the Arab American community. Every session began with a “wise old Arabic saying” concerning goal attainment. Imagery incorporating cultural themes was readily used, such as describing food as medicine for the body and soul. Demonstrations were also culturally-targeted, with interventionists modifying traditional Arab cooking techniques and encouraging participants to discuss the feasibility of incorporating these changes in their family meals.

In contrast with the DPP, this translated intervention delivered sessions via groups of 10–12 individuals. The investigators decided on a group format for two primary reasons. First, they identified social interactions as an important factor for this population and wanted to preserve as much of that cultural value as was possible. Second, family is perhaps the most central value for Arab Americans;

family support was strongly encouraged, with non-participants allowed to attend group sessions. Notably, same-gender groups were requested by approximately 20 % of participants.

Intervention content was modified to address cultural factors influencing diet and physical activity. Regarding diet, simple modifications were emphasized and individualized for each participant based on their weight. Culturally-specific foods were substituted into the “DPP Lifestyle Balance Fat Counter” to make the handout more relevant to participants. Physical activity demonstrations were held in gender-segregated groups, with emphasis on walking in groups and aerobic classes, as these were commonly endorsed forms of exercise among participants. Finally, written intervention materials were provided in both English and Arabic.

Bilingual study personnel were used to deliver all study procedures. These individuals identified as being of Arabic descent and were thoroughly trained by the study investigators and medical personnel. Intervention fidelity was assessed constantly, both through observation and review pre- and post-sessions.

Not all eligible participants were initially interested in participating in the modified DPP protocol. These individuals were invited to attend a four-session seminar. Following the Health Belief Model [49], the investigators developed an educational intervention to target ambivalent participants’ beliefs about diabetes regarding: perceived susceptibility, perceived severity, perceived benefits, perceived barriers, and self-efficacy. Following the seminar, those interested were invited to enroll in the lifestyle behaviors intervention.

Success of Translation Methods

The study indicated a high level of acceptability among participants. Adherence to the intervention was high, with 92 % of participants attending at least 10 weekly sessions. Retention throughout the 24-week protocol was also high (86 %). Outcome measures indicated significant reductions in total and saturated fat intake, and three out of four participants achieved the 150-min physical activity goal.

The “pre-intervention” educational intervention for ambivalent participants was also successful. Forty-nine out of 63 participants of these participants ultimately converted and were motivated to enroll in the lifestyle intervention.

This modified protocol appropriately targeted a community at risk of T2D and developed a culturally-adapted translation of the DPP to suit the community’s self-identified health needs. The investigators demonstrated the effectiveness of the program, and creatively problem-solved a way to retain initially-ambivalent participants (with the educational intervention). Moreover, the authors made diet and physical

activity modifications that were culturally-sensitive. The authors used bilingual study personnel who were part of the community, characteristics that may create added trust and thus acceptance of intervention material [50].

Alternative or Supplemental Methodologies for Future Translation Efforts

The authors solicited input from the community before developing intervention materials, which likely maximized the potency of their modified protocol. However, with no control group utilized, it is unclear if the extent to which their effort in soliciting this input was warranted. Moreover, the extent of the accommodations the authors provided (i.e., using an education intervention, adding a session for Ramadan) may be difficult for less-knowledgeable interventionists to implement in other Arab American communities. This study relied on Arab American study personnel; in healthcare or community clinics, the intervention deliverer may not be an Arab American, so this aspect of their protocol may be unfeasible in future application.

American Indian and Native Alaskan Translation

The risk of developing T2D is high in American Indian and Native Alaskan (AI/NA) populations, with over 16 % of adults currently diagnosed and an additional 20 % with prediabetes [13]. Unfortunately, these high prevalence rates may only increase; T2D incidence rates among AI/NA aged 10–19 years old is above 35 %, and is the highest among any ethnic community in the United States in that age range [11]. Charged with addressing this major public health concern by Congress, the Indian Health Service (IHS) designed and implemented the Special Diabetes Program for Indians Diabetes Prevention (SDPI-DP), a DPP translation project that began development in 2004, initiated recruitment in 2008, and recently published its findings [26]. This wide scale study examined the translational effectiveness of the DPP across 36 AI/AN health care programs, which included rural, reservation and urban AI/NA communities. Jiang et al. [26] reported on data from 2553 AI/NA participants. Each of the 36 programs included in the study was required to implement the 16 session Life Balance Curriculum from the DPP in group settings in a 16–24 week time period. Following the 16 session curriculum, monthly individual lifestyle coaching sessions were offered to participants to help with goal setting and problem-solving. Participants were followed for up to 3 years for the purpose of assessment. Due to strong evidence for the efficacy of this lifestyle approach to diabetes prevention, the inclusion of a waitlist-control group was deemed unethical and not included in the study.

Translation Methods Utilized

Of the 36 programs involved in the SDPI-DP, 6 were IHS hospitals or clinics, and 30 were tribal health care programs or IHS-contracted health care programs that were administered by tribes. The partnership between the programs and IHS began with a year of bi-monthly meetings, which allowed the programs to become oriented to the DPP and to come together to develop activities. The year also allowed the programs time to effectively market their efforts to their communities and complete local approval processes. Principles of community-based participatory research [51] were embraced in this process, and allowed for health program staff to assist in the development of evaluations. Jiang et al. [26] reported that staff members were encouraged to provide guidance on the acceptability of evaluation designs within their respective tribes in order to help prioritize the factors that would promote the intervention's success at the program and individual level.

Programs primarily recruited participants through community events, such as health fairs. Participants were also recruited directly from local clinics or by provider referral. These recruitment methods allowed for potential participants to be drawn exclusively from the AI/AN community and allowed for confirmation of the participants AI/AN identification (determined by eligibility for IHS services). While each program was required to cover the basic 16-lesson DPP curriculum, adaptations to the curriculum for local cultures was allowed. For example, many of the health care programs involved in the study chose to translate concepts and the curriculum into their tribal languages. AI/AN traditions, such as talking circles, indigenous foods, and drumming, were also incorporated into many of the sessions.

Success of Translation Methods

On average, participants attended 13/16 classes, indicating that the program was well received by the participants. The annual crude incidence of diabetes for participants who finished all 16 classes was substantially lower (at 3.5 % per year) than participants who did not finish all of the classes (at 7.5 % per year). Participants who attended all of the classes also had larger reductions in fasting blood glucose level, weight, and systolic blood pressure at the post-treatment assessment. Post-intervention, participants reported significantly higher levels of physical activity (~181 min/week) compared to baseline (~99 min/week). Physical activity habits were largely sustained at long term follow-up (~150–160 min/week). Fasting blood glucose levels also decreased among participants from baseline to all follow-up assessments. The average weight loss from baseline to post-treatment assessment was 9.6 lbs (4.4 % weight loss).

Alternative or Supplemental Methodologies for Future Translation Efforts

To date, this study has been the largest DPP translation effort to an ethnic minority group in the United States, and it was largely successful. Because it was deemed unethical to include a control condition, it is difficult to ascertain the comparative effectiveness of lifestyle modification to no treatment in this American subpopulation. For future translation efforts, an additional focus should be on developing effective retention strategies. This study was plagued by retention difficulties, and the authors do not note the retention strategies utilized by the health care programs involved. Future efforts may utilize retention strategies known to be effective for other minority groups, such as collecting contact information for other family members, and sending text messages and e-mails to contact participants who may be hesitant to answer the telephone [52]. Further, the authors noted difficulty in forming groups of 8–12 participants due to the varying schedules of participants in the program. The average group size was 4 participants per class. Because group-based lifestyle interventions are more cost-effective than one-on-one sessions [53], it is important to establish and maintain an adequate class size. Future translational efforts may consider the utilization of videoconferencing as a method for treatment delivery which could help to unite small groups from the same tribes together, and while this would require close communication and organization between health care programs, behavioral interventions delivered via videoconferencing technology have been effective in a variety of groups [54]. This may offer cost-savings for health care providers, as well as offer additional social support to participants.

Summary and Future Directions

In the United States, African American, Hispanic/Latino, Native Hawaiian and Other Pacific Islander, Arab American, and American Indian and Native Alaskan communities are disproportionately at risk of developing T2D. The DPP has provided researchers interested in reducing these health disparities with a solid foundation for a lifestyle intervention, targeting diet and physical activity. This review offered a critique of translation efforts targeted at each of these groups. There were several desirable features that characterized these endeavors. First, the concerted effort to obtain input and participation from community leaders and members is a strength [22, 24–27] and has the capacity to add practical components that may be culturally relevant. Second, while the DPP initial 16-week core lifestyle intervention was individually administered, group-based delivery was common to the translation programs

reviewed. The group-based approach is a significant strength as it is cost effective and is a vehicle that taps into the high value that certain ethnic minorities may place on community, family, and collectivism [25, 53]. Furthermore, the inclusion of bilingual staff and materials, features of the work with Hispanics/Latinos and Arab Americans, reduces barriers to treatment, and may make participants feel as though the intervention is personalized, a factor which can influence intervention effectiveness [32, 55]. This strongly suggests that future translations of the DPP be responsive and sensitive to the needs of non-native English speakers.

Despite the notable strengths, there is considerable room for improvement in future efforts. For example, the faith-based focus of the work with African Americans capitalizes on past health promotion efforts but the absence of culturally relevant lifestyle content and recommendations warrants further attention [22, 27]. At a minimum, any work with ethnic communities should consider cultural traditions and include healthy alternatives and/or modifications to traditional recipes and employ culturally accepted options and formats for physical activity [25]. In addition, physical activities and other forms of health behavior change that incorporate the presence of others and provide social support may increase the likelihood of adherence to activity recommendations [33, 56]. Inclusion of at least one theoretically-based pretreatment session should be routinely implemented as it has the potential to engage patients, provide a strong treatment rationale, and maximize adherence and retention [25]. Furthermore, the reach of ethnic translations of the DPP needs to be broadened especially with regard to Asian Americans, who are at high risk for T2D [57].

To facilitate dissemination in community settings, it would be advisable to determine the sufficient and optimal dose of the intervention. For example, the lengthy intervention of the DPP and the adaptation for SDPI-DP [26], would not be feasible in the average health care setting. With the transition from efficacy to effectiveness programs, attention needs to be directed at how many sessions are sufficient to obtain the desired outcomes and what is feasible in clinical practice.

The reviewed studies have all been recently published, underscoring the nascence of national attention given to developing culturally informed translations of efficacious T2D preventions. As indicated in this review's introduction, several other major lifestyle interventions have established efficacy. The relative success of the DPP's translation efforts may indicate similar effectiveness of these protocols. Ethnic translations of successful interventions, such as the DPP, are critical for successful dissemination to and utilization by populations in need. As the United States and other industrialized countries increase in

diversity, healthcare disparities research, and the interventions they produce, should continue to be a priority of health reform efforts nationwide.

Conflict of interest The authors have no conflict of interest to declare.

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